

Reverse Engineering a High-Temperature Heat Shield for Interstellar Propulsion Using Blue Light Scanning: Applications of 3D Scanning in Polygonal Geometry

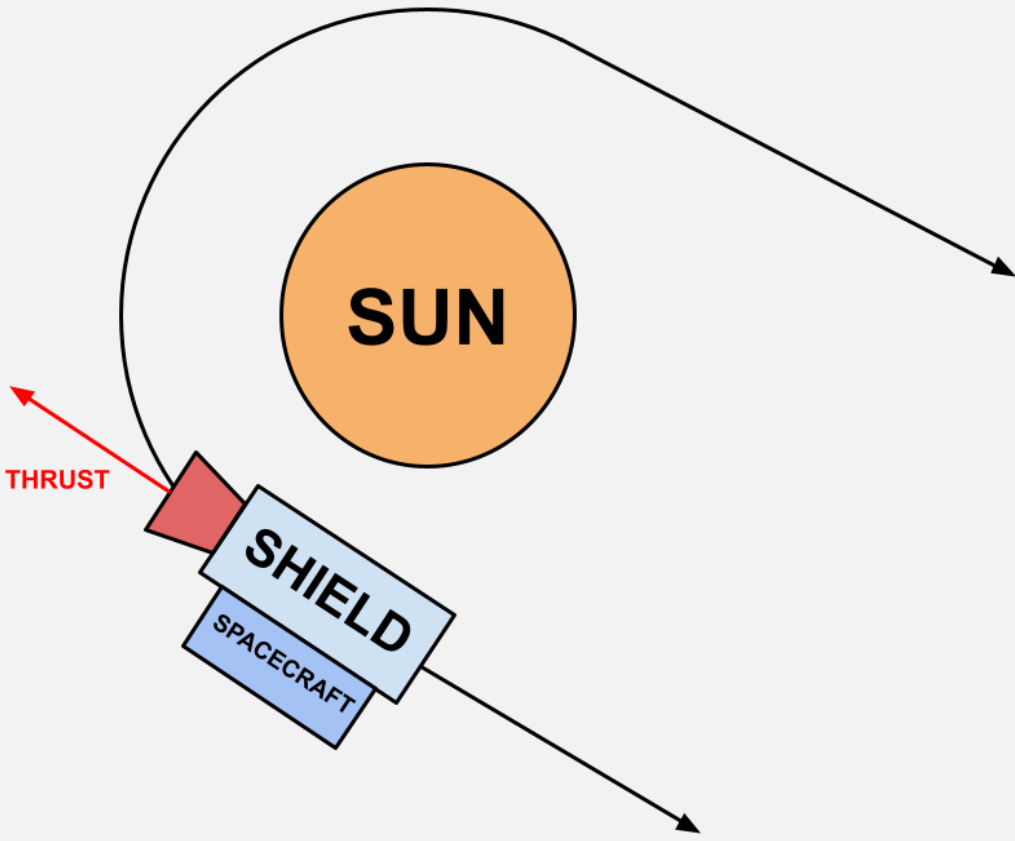


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What is Solar Thermal Expansion?

The principle behind solar thermal propulsion is to create thrust by using energy from the sun to heat a fluid propellant, causing it to expand and generate force when routed through a nozzle. Proximity to the sun is limited by the temperature that a spacecraft can survive, so heat exchangers are necessary to provide spacecraft with high specific impulse while simultaneously cooling the system.



In high-temperature aerospace environments, heat radiation can lead to significant warping and deformation of objects. To address this, advanced metrology tools, such as ATOS Professional, are relied upon. When paired with a structured blue light scanner, ATOS Professional allows for precise capture of post-test geometries, enabling direct comparison with original CAD models or pre-test scans.

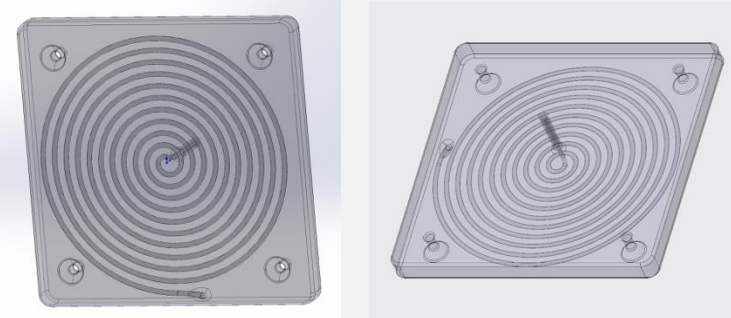
Equipment/Methodology

Software:

- Geomagic™ Design X
- ATOS Professional by Zeiss®

Materials:

- *.SLDPRT file of heat shield design
- ATOS Scanner
- Sticker nodes
- Rotating table
- Heat shield
- 4 foam balls



To capture the geometry, the heat shield was mounted onto a lazy Susan by making a pyramid with foam balls. Sticker nodes were then placed onto the table, as these help the scanner identify and calibrate the proportions of the scanned object. Several scans were then taken from multiple angles by controlling the camera remotely from a separate computer. The resulting point cloud data was then observed and analyzed, then compared with the original CAD model.

Using the ATOS camera to scan the heat shield in the ideal setup.

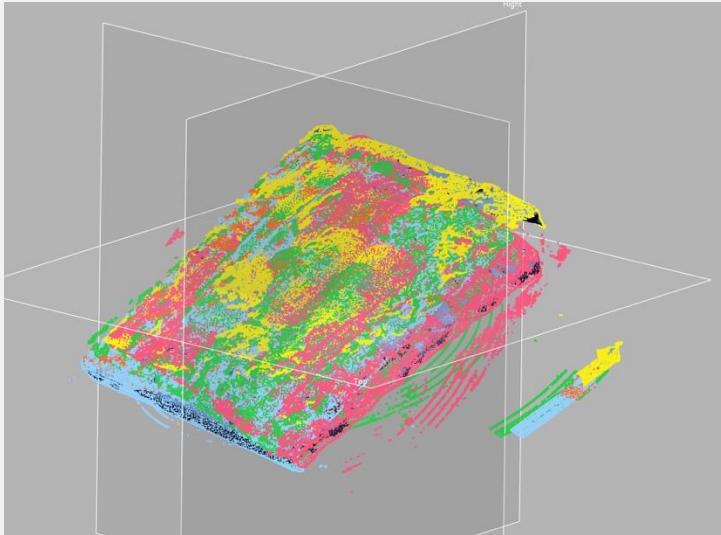


Point Cloud Results

The GOM blue light laser scanner produced a high-resolution point cloud of the heat shield's post-test geometry. Upon analysis, noticeable deformation patterns were observed.

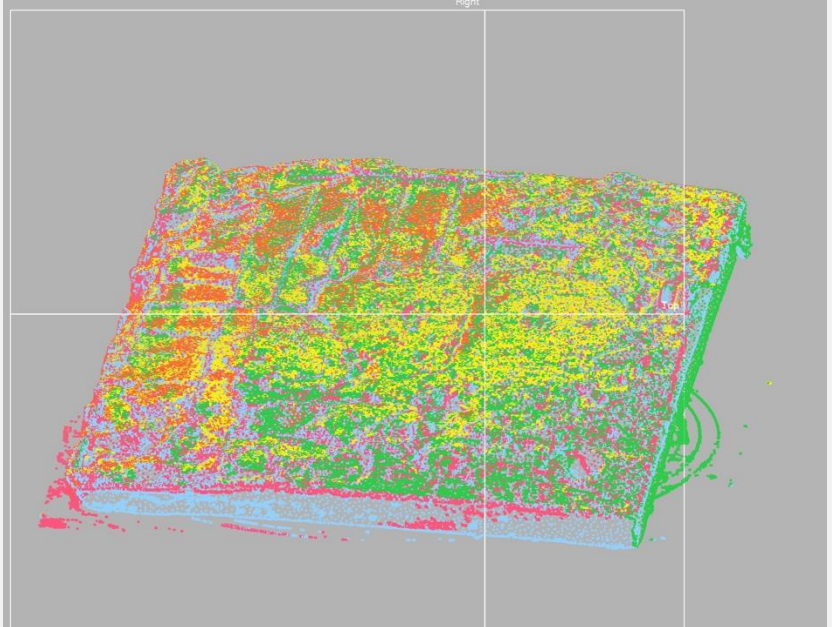
These included:

- **Warping**, particularly along the edges and thin sections
- **Localized volume loss**, most prominent near the central heat-exposed region
- **Surface deviation**, with displacement measurements quantifying how far the geometry shifted from its original state



Red and yellow are indicative of high deviation from the original geometry.

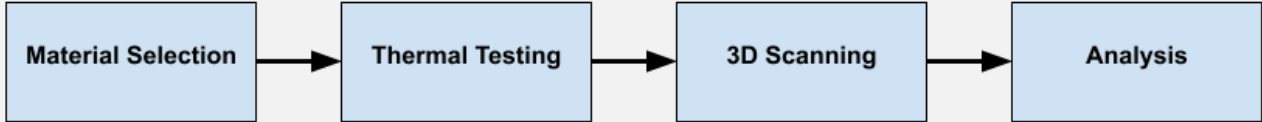
Color variation indicates areas of material shift and volume loss due to heat-induced stress.



Conclusion and Next Steps

The results clearly showed that heat radiation led to measurable warping, particularly along the edges of the panel, with surface deviations and volume loss indicating material fatigue and structural compromise. These findings highlight the importance of integrating precise 3D metrology into the evaluation process for thermal protection systems.

Moving forward, further testing should be conducted across a broader range of materials and thermal conditions to validate durability over repeated exposures. Ultimately, this process will support the design of more resilient heat shields for future aerospace applications.



Acknowledgements

I would like to thank my wonderful mentor, Mr. Michael Boyle, for providing me with the connections and resources necessary to perform this research project, guiding me through the research process, and devoting hours of his time to help me. I would also like to thank Mr. Kurt Ruckelshaus for sharing his expertise on 3D scanning and Ms. Jessica Harsono for providing some relevant background information regarding the construction and implementation of the heat exchanger. Lastly, thank you to the ASPIRE program and the Johns Hopkins University Applied Physics Laboratory for providing me with such an invaluable experience!